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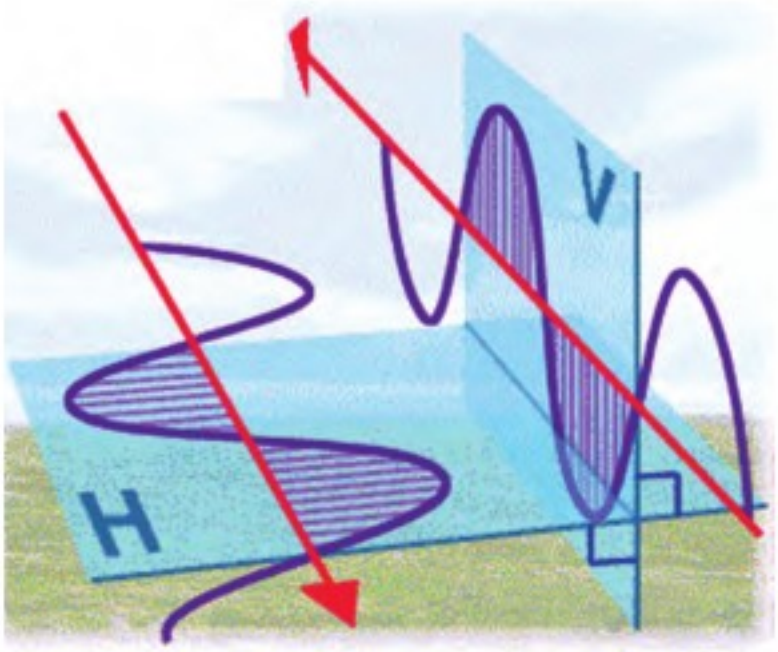
CHAPTER



Polarization

20.1. INTRODUCTION

Interference and diffraction phenomena proved that light is a wave motion and enabled the determination of the wavelength. However, they do not give any indication regarding the character of the waves. Whether the light waves are longitudinal or transverse, or whether the vibrations are linear or circular cannot be deduced from the above two phenomena, as all kinds of waves under suitable conditions exhibit interference and diffraction. In 1816 Arago and Fresnel showed that light waves vibrating in mutually perpendicular planes do not interfere. In 1817 Thomas Young postulated that light waves are *transverse waves* and explained



Polarization

At a Glance

- Introduction
- Unpolarized and Polarized Light
- Types of Polarization
- Polaroid Sheets
- Malus' Law
- Double Refraction in Calcite Crystal
- Effect of Polarizer on Light of Different Polarizations
- Experimental Determination of Principal Refractive Indices
- Phase Difference Between E-ray and O-ray
- Retarders
- Production of Circularly Polarized Light
- Babinet Compensator
- Double Image Polarizing Prisms
- Optical Activity
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- Electro-optic and Magneto-optic Effects
- Magneto-optic Effects
- Anisotropy Induced by Mechanical Strain
- Photoelasticity

the absence of interference between light waves polarized in mutually perpendicular planes. Thus, the existence of polarization property is a direct consequence of light being a transverse wave. Light coming from common light sources is unpolarized. It can be transformed into different types of polarization using optical devices. The state of polarization cannot be perceived by an unaided human eye. An understanding of polarization is essential for understanding the propagation of electromagnetic waves guided through wave-guides and optical fibres. Polarized light has many important applications in industry and engineering. One of the most important applications is in liquid crystal displays (LCDs), which are widely used in wristwatches, calculators, TV screens, etc.

20.2. POLARIZATION

Waves are basically of two types: (*i*) longitudinal waves and (*ii*) transverse waves.

(*i*) A wave in which particles of the medium oscillate to and fro along the direction of propagation is called a **longitudinal wave**. Waves produced on a spring and sound waves are examples of longitudinal waves. The longitudinal wave consists of alternate compressions and rarefactions, as shown in Fig.20.1 (c) & (d).

(*ii*) A wave in which every particle of the medium oscillates up and down at right angles to the direction of wave propagation is called a **transverse wave**. Ripples on water surface and waves on a rope are examples of transverse waves. The wave propagates in the form of alternating crests and troughs, as shown in Fig.20.1 (a) & (b).

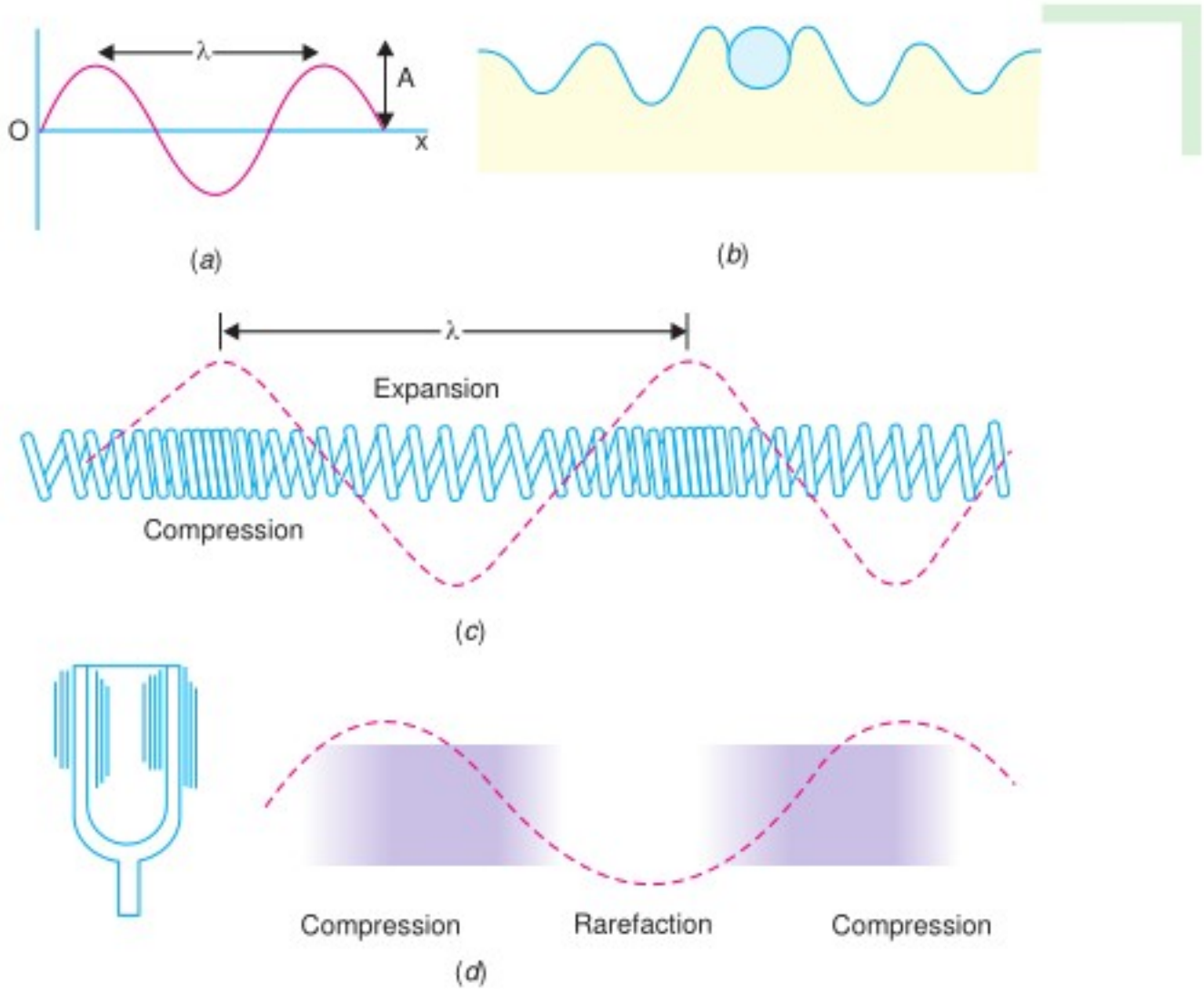


Fig. 20.1

In a longitudinal wave, all directions perpendicular to the wave propagation are equivalent. On the other hand, a preferential direction normal to the wave propagation exists in a transverse wave.

The preferential direction in a transverse wave is the direction of vibration of the particles and it differs from all other directions. The existence of a preferential direction for a transverse wave leads to the characteristic phenomenon known as polarization. Polarization is not found with longitudinal waves as they do not possess a directional property. Thus, polarization is specific to transverse waves.

Light waves are transverse waves consisting of electric and magnetic fields vibrating perpendicular to each other and to the direction of propagation. The vibrating electric field vector and the direction of propagation of the wave constitute a plane. There is an infinite number of such planes around the direction of propagation. In an ideal light wave, the vibrations of electric vector are confined to a single plane. In practice, light sources emit a mixture of light waves whose planes of vibration are randomly oriented about the direction of propagation. Such random orientation of vibration planes gives rise to symmetry about the wave propagation direction. As a result, the transverse nature of the wave gets concealed. The process of removing the symmetry and bringing in one-sidedness in the light wave is called **polarization**.

20.3. UNPOLARIZED AND POLARIZED LIGHT

Light wave is a transverse electromagnetic wave made up of mutually perpendicular, fluctuating electric and magnetic fields. Fig.20.2 (*a*) shows the electric field in the *xy*- plane, the magnetic field in the *xz*-plane and the propagation of the wave in the *x*- direction. The right-hand part of the diagram shows the variation of the electric field in space as the wave propagates. Traditionally, light wave is described by the **electric field vector**, ***E***, and accordingly, only the electric field vector is shown in Fig.20.2 (*b*).

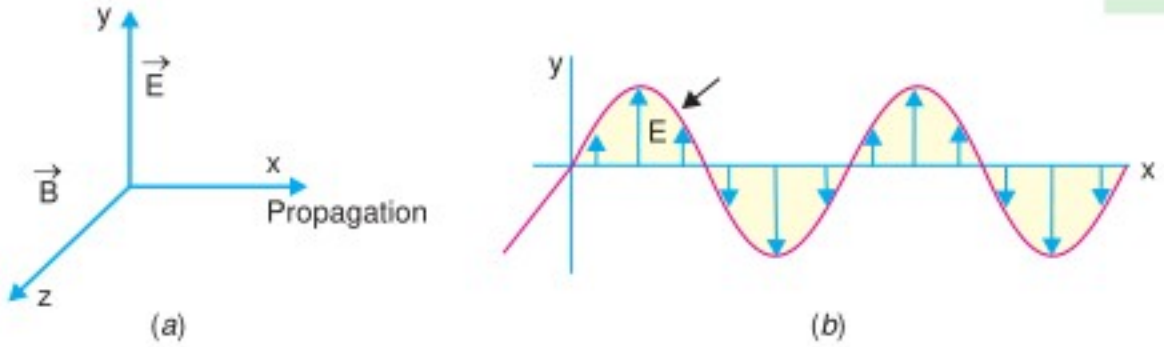


Fig. 20.2: A light wave is described by the electric field vector, *E*

As the electric field is a vector, it points in a particular direction in space. The **polarization** of an electromagnetic wave refers to the orientation of its electric field vector *E*. If we could view a light wave coming from an ordinary source towards us, then we would observe that the direction of *E* is randomly varying with time on a very fast scale. The light from an incandescent bulb, for example, emits a mixture of light waves with electric field components that change randomly on a scale of 10^{-14} s, almost as fast as the optical frequency itself. As a result, the **direction of oscillation of the electric field vector in an ordinary light beam occurs in all the possible planes perpendicular to the beam direction,** as illustrated in Fig.20.3. A light wave, in which **E**-vector oscillates in more than one plane, is referred to as **unpolarized light**. Light emitted by the sun, by an incandescent lamp, or by a candle flame is unpolarized light.

Polarized Light

Polarized light is not produced naturally. It is obtained by converting natural light into polarized light using optical elements. The process of transforming unpolarized light into polarized light is polarization. A polarized light wave is a light wave with a definite direction of oscillation of the *E*- vector, which occurs in a *single plane* or in *some specific way*. For example, the wave in Fig.20.2 (b) is a polarized wave. **Polarized light** is the light that contains waves that only fluctuate in one specific plane.

We designate the plane created by the direction of oscillation of the electric field vector $\boldsymbol{E}$ and the direction of propagation of the beam as the **plane of polarization of light wave**. Thus, the *xy*-plane is the plane of polarization in Fig.20.2 (*b*).

20.4. NATURAL LIGHT IS UNPOLARIZED LIGHT

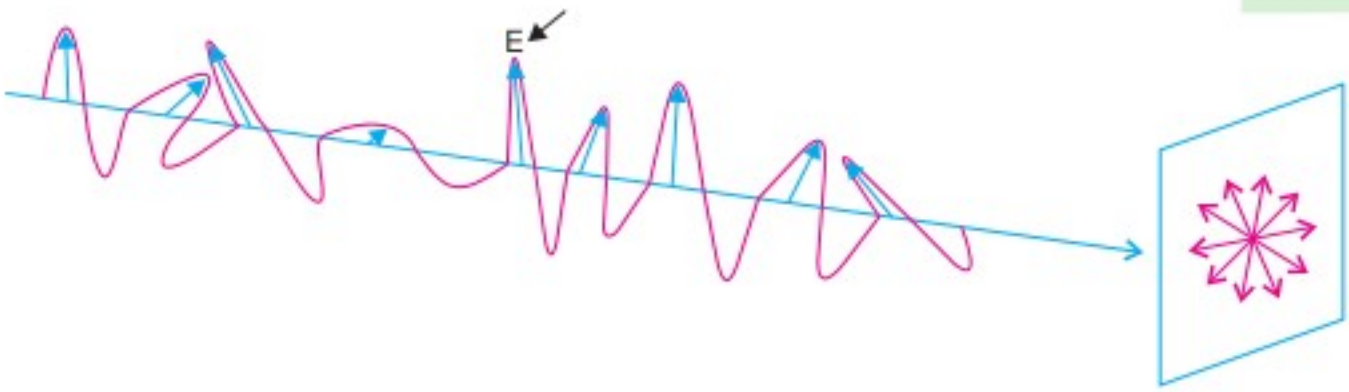


Fig. 20.3: Natural light is unpolarized

We mentioned earlier that light from the ordinary light sources is unpolarized. Let us find out the reason. We know that atoms emit light. Any light source consists of a very large number of atomic emitters. Each atom radiates, at a specific instant, a **wave packet** (also known as a **wave train**) that lasts for about 10^{-8} s. Light radiated by a source is a mixture of wave packets emitted by different atoms at different instances (see Fig.20.3). Individual wave packets will be polarized, but each wave packet has its own polarization, and they are not correlated in any way. There is no continuity of plane of polarization and the plane of polarization varies from wave packet to wave packet in a completely random manner. The polarization of the waves determined at any particular spot will fluctuate randomly, and very rapidly, with no preferred direction. Any direction is equally likely, and the usual graphical representation of ordinary light is shown in the right-hand figure in Fig.20.4 (*a*). Fig.20.4 (*a*) is a fictitious diagram. It only implies that the natural light consists of electric field vectors of many possible orientations lying at different angles between 0 and 360° and hence symmetrically distributed about the direction of propagation.

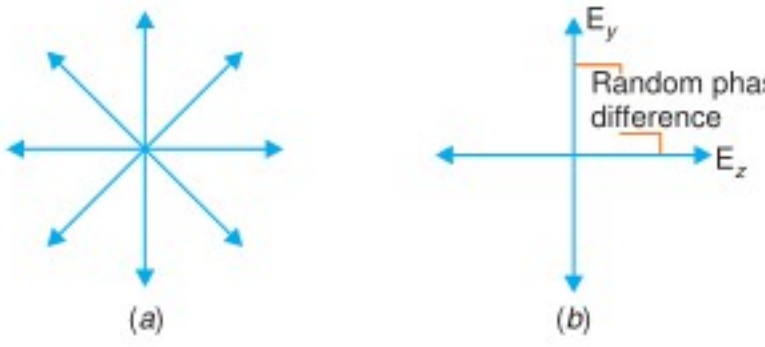


Fig.20.4: (*a*) Pictorial representation of Natural light. (*b*) Unpolarized light is viewed as a combination of *incoherent* vertically and horizontally polarized waves.

The concept of unpolarized light is rather difficult to visualize. In general, it is looked upon as consisting of an average of half of its vibrations horizontally polarized and half of its vibrations vertically polarized. Due to the random distribution of the optical vectors, the amplitude of the vertical and horizontal component vectors are taken as equal; however, the **two components are incoherent**, *i.e.* have a randomly changing phase difference. In view of this, unpolarized is pictorially represented by two electric vectors as shown in Fig.20.4 (*b*).

A comparison of Unpolarized and Polarized light

Unpolarized light	Polarized light
1. Consists of waves with planes of vibration equally distributed in all directions about the ray direction.	Consists of waves having their electric vector vibrating in a single plane normal to ray direction.
2. Symmetrical about the ray direction	Asymmetrical about the ray direction.
3. Produced by conventional light sources.	Is to be obtained from unpolarized light with the help of polarizers.
4. May be regarded as the resultant of two *incoherent* waves of equal intensity but polarized in mutually perpendicular planes.	May be regarded as the resultant of two mutually perpendicular *coherent* waves having zero phase difference.

20.5. TYPES OF POLARIZATION

The polarization of a light wave describes *the shape and locus of the tip of the E vector* (in the plane perpendicular to the direction of propagation) *at a given point in space as a function of time*. Depending upon the locus of the tip of the E vector, light may exhibit three different states of polarization. They are

(*i*) plane or linear polarization,

(*ii*) elliptical polarization and

(*iii*) circular polarization.

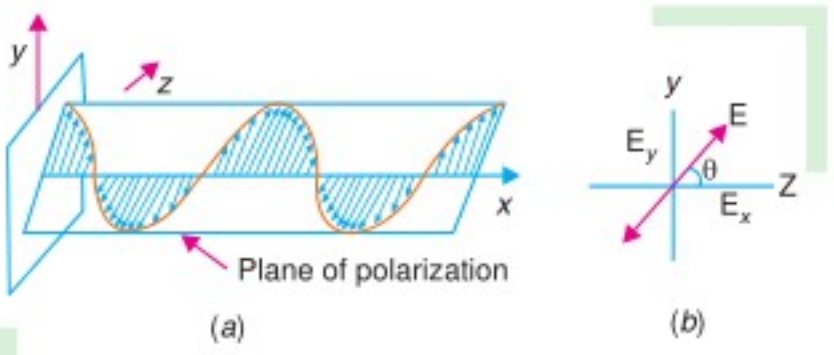
Apart from these, the light may also be partially polarized.

An unaided human eye cannot identify the state of polarization of light. However, some insects and animals possess polarization sensitive vision.

20.5.1 PLANE POLARIZED LIGHT

Plane polarized light waves are light waves in which the oscillations occur in a single plane. In a plane-polarized wave, the oscillations of electric field vector E are strictly confined to a single plane *perpendicular* to the direction of propagation. As the direction of the field vector at some point in space and time lies along a line in a plane perpendicular to the direction of wave propagation, a plane-polarized wave is also known as a **linearly polarized** wave.

With linear polarization, the orientation of the E-vector stays constant at a point in space. That is, the direction of E does not vary with time, but its magnitude varies sinusoidally with time. If the field is pointing either up or down, we call it **vertical polarization**, and if it's pointing either right or left, we call it **horizontal polarization**. Electric fields are not restricted to pointing exactly along vertical or horizontal axes, but can be at any arbitrary angle to those axes. Linearly polarized light, polarized at any arbitrary angle, may be regarded as a combination of horizontally and vertically polarized light, with appropriate amplitude, and which are oscillating **in phase** or **180° out**

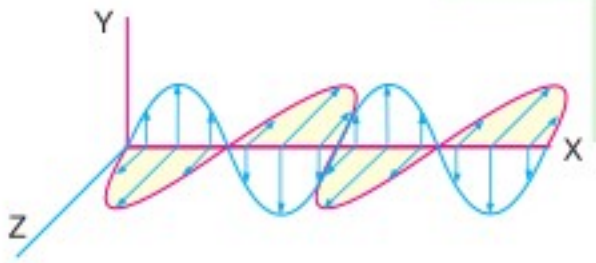


A light wave polarized in an arbitrary direction

Fig.20.5

of phase. The **key point** is that the two component waves are **coherent**.

Let ***x*** be the direction of travel of the light and, *y* and *z* be directions in the plane of the electric field. The electric field makes a constant angle to the *z*-direction, as shown in the Fig.20.5 (*a*). The wave in Fig.20.5 (*a*) is the resultant wave due to superposition of two **coherent** linearly polarized waves, oscillating in phase, as shown in Fig.20.6.

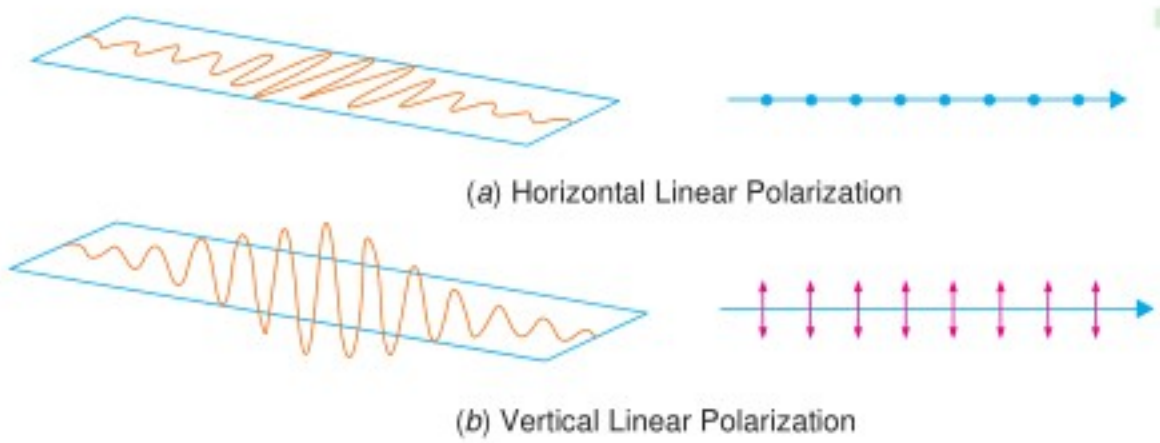


A linearly polarized wave is viewed as a combination of *coherent* vertically and horizontally polarized waves.

Fig.20.6

Representation of linearly polarized light in diagrams

Linearly polarized light is represented in diagrams as shown in Fig.20.7. When the electric field vector oscillates horizontally in a direction perpendicular to the plane of the paper, the light wave is represented by dots (Fig.20.7 *a*). When the electric field vector oscillates vertically in the plane of the paper, the light wave is represented by arrows, as shown in Fig.20.7 (*b*).



(*a*) Horizontal Linear Polarization

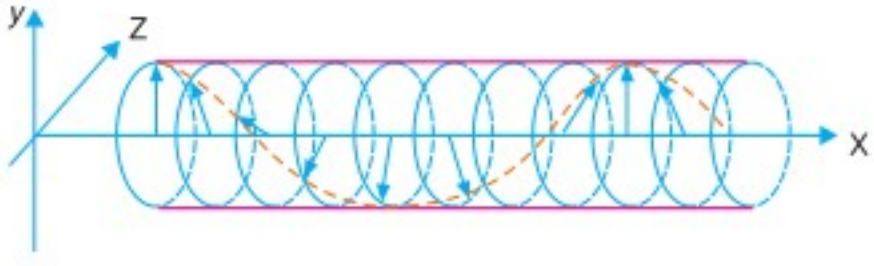
(*b*) Vertical Linear Polarization

Representation of linearly polarized light in diagrams

Fig. 20.7

20.5.2 CIRCULARLY POLARIZED LIGHT

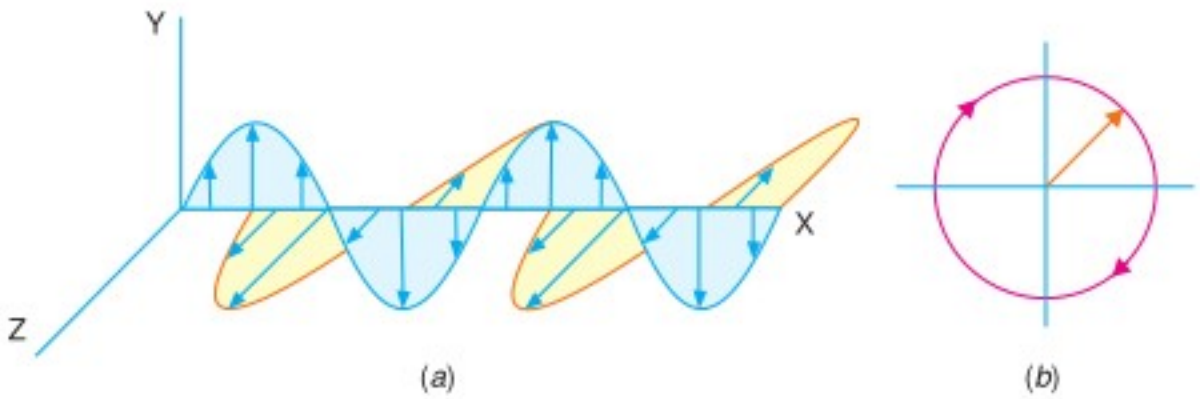
A light wave is said to be ***circularly polarized***, if in the course of wave propagation, the magnitude of the electric vector *E* stays constant but it rotates at a constant rate about the direction of propagation and sweeps a circular helix in space, as shown in Fig.20.8. This is a picture in terms of the **space variation** of *E*. Alternately, if we could see the wave advancing towards our eyes, we would find that the tip of the *E* vector tracing a circle in space, completing one revolution within one wavelength. Hence, the state of polarization is called *circular polarization*. In, circularly polarized light, there is no preference to specific direction of oscillation.



Space variation of *E*; *E* vector sweeps a circular helix in space

Fig. 20.8

A circularly polarized light wave may be regarded as the resultant wave produced due to superposition of two coherent linearly polarized waves of *equal amplitude* oscillating in mutually perpendicular planes, and are out of phase by 90°.

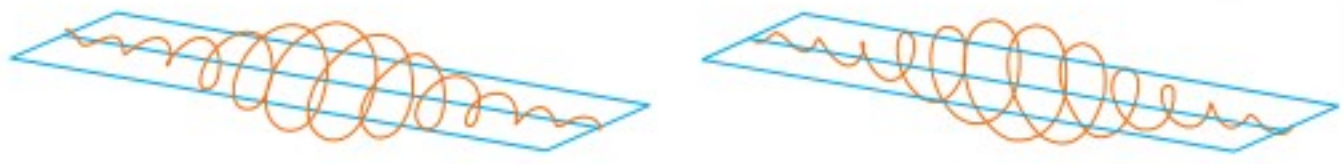


Circularly polarized wave is a combination of horizontally polarized and vertically polarized waves that are out of phase by 90° and having equal amplitudes.

Fig. 20.9

Let us again consider two linearly polarized waves having equal amplitude, out of which one is polarized in *z*-direction (horizontally polarized wave) and the other in *y*-direction (vertically polarized wave). Let us further assume that they are **coherent** and out of phase by 90° (see Fig.20.9a). At some instant, the *E*-field vector of *z*-polarized wave will have maximum amplitude and the *E*-field vector of y-polarized wave will be at zero. At that instant, the polarization is horizontal. A little time later, the *z*-polarized wave has decreased a little, while the *y*-polarized wave has started to increase. Then, the light looks like it is polarized at a slight angle— mostly horizontal, but with a small vertical component. A little while later, the *z*-component has decreased some more, and the *y*-component has increased some more, and the angle is greater. And so on. Eventually, the *y*-polarized wave is at a maximum, and the *z*-polarized wave is zero, and we have pure vertical polarization. If we stand at one point in space, and look at the direction of the wave, we will observe that the *E*-vector sweeps a circle in space. Hence, it is called circularly polarized wave. Note that the *oscillations of the resultant E-vector do not take place in a single plane.*

If the rotation of the tip of *E* is clockwise as seen by an observer looking back towards the source, then the wave is said to be **right-circularly polarized** (Fig.20.10 *a*). If the tip of *E* rotates anti-clockwise, then the wave is said to be **left-circularly polarized** (Fig.20.10. *b*). This is a description of circular polarization in terms of the **time variation** of *E*.



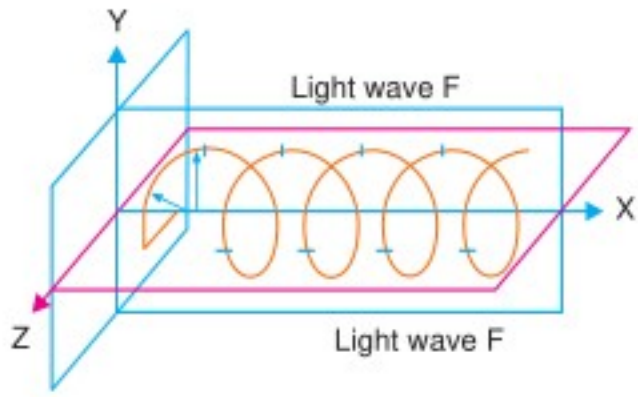
(a) Right circular polarization (b) Left circular polarization

Fig. 20.10

20.5.3 ELLIPTICALLY POLARIZED LIGHT

A light wave is said to be ***elliptically polarized***, if the magnitude of electric vector *E* changes with time and the vector *E* rotates about the direction of propagation and sweeps a flattened helix in space, as shown in Fig.20.11. This is a description of elliptically polarized light in terms of the **space variation** of ***E***. Alternately, if we imagine that we are looking at the light wave advancing towards us, we would observe that the tip of the *E* vector traces an ellipse in space. Hence, it is called elliptically

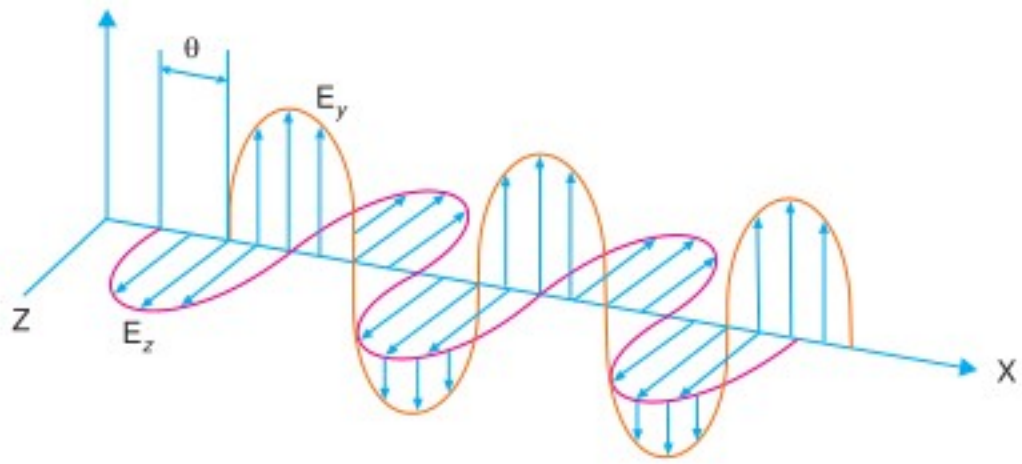
polarized light. This is a description of elliptical polarization in terms of the **time variation** of *E*.



Space variation of **E**; **E** vector sweeps a flattened helix in space

Fig. 20.11

An elliptically polarized light wave may be regarded as the resultant wave produced due to superposition of two **coherent** linearly polarized waves of different amplitudes, oscillating in mutually perpendicular planes and are out of phase. If waves of differing amplitude are related in phase by 90°, or if the relative phase difference is other than 90° then the resultant light wave is elliptically polarized.



Elliptically polarized wave is a combination of horizontally polarized and vertically polarized waves that are of different amplitudes and out of phase by θ

Fig. 20.12

Let us consider two linearly polarized waves having different amplitudes, out of which one is polarized in *x*-direction and the other in y-direction. Let us further assume that they are **coherent** and out of phase by an arbitrary angle θ (see Fig.20.12). The *E*-field vector of *Z*-polarized wave will have maximum amplitude at times when the *E*-field vector of *y*-polarized wave is a minimum and vice versa (Fig.20.12). Then the oscillations of the resultant *E*-vector do not take place in a single plane. The magnitude of resultant *E*-vector varies at each point in space and the overall rotation of the *E*-vector has the appearance of a flattened helix.

When we are looking back towards the source, if the rotation of *E* vector occurs clockwise, it is said to be a **right-elliptically-polarized** wave. If it rotates anti-clockwise, it is said to be a **left-elliptically polarized** wave.

20.5.4 PARTIALLY POLARIZED LIGHT

Usually, light is neither totally polarized nor unpolarized but a mixture of the two types. It can be viewed as a mixture of plane polarized light and unpolarized light. Partially polarized light is represented as shown in Fig.20.13.

Partially polarized light, like natural light, can be represented in the form of a superposition of two incoherent plane-polarized waves with mutually perpendicular planes of oscillations. In case of natural light the amplitude of these waves is the same and for partially polarized light, it is different.

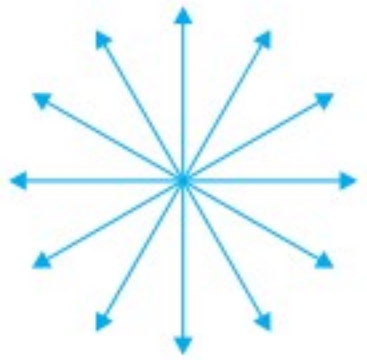


Fig. 20.13

Degree of Polarization:

If we pass partly polarized light through a polarizer, and if we rotate the polarizer about the direction of the ray, the intensity of the transmitted light will change within the limits from I_{max} to $I_{min.}$. The transition from one of these values to the other will occur upon rotation through an angle of 90°. We define the degree of polarization with the help of the following expression.

$$P = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} \quad \text{... (20.1)}$$

% polarisation

$$\% \text{ polarization} = \frac{(I_{max} - I_{min})}{(I_{max} + I_{min})} \times 100 \quad \text{... (20.2)}$$

For plane polarized light $I_{min} = 0$, and hence $P = 1$ and the % polarization is 100%. For natural light, $I_{max} = I_{min}$, and hence $P = 0$ and the % polarization is zero. If $I_{max} = 2I_{min}$, $P = 0.33$ and % polarisation = 100/3 = 33%.

The concept of the degree of polarization cannot be applied to elliptically and circularly polarized light.

20.6. PRODUCTION OF PLANE POLARIZED LIGHT

We now study the methods of producing plane-polarized light. Plane polarized light may be produced from unpolarized light using the following five optical phenomena:

(*i*) reflection, (*ii*) refraction, (*iii*) scattering, (*iv*) selective absorption (dichroism), and (*v*) double refraction. Out of these five, the phenomena of selective absorption and double refraction are helpful in practical production of plane polarized light.

20.6.1 POLARIZATION BY REFLECTION FROM DIELECTRIC SURFACES

E.L.Malus, French engineer discovered in 1808 polarization of light by reflection. He noticed that when natural light is incident on a smooth surface, at a certain angle the reflected beam is plane polarized. The extent to which polarization occurs is dependent upon the angle at which the light is incident on the surface and upon the material, which the surface is made of. Metallic surfaces reflect light with a variety of vibrational directions; such reflected light is unpolarized. However, light that is specularly reflected from dielectric surfaces, such as asphalt roadways, water etc, is linearly polarized. If the extent of linear polarization is large, a person perceives **glare** from such surfaces. On bright sunny days, the glare caused by sunlight on a roadway or a field of snow, may be almost blinding to the human eye.

When light wave is incident on a boundary between two dielectric materials, part of it is reflected, and part of it is transmitted.